

ROHIT BANGAL

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ELECTRICAL ENGINEER WITH EXPERTISE IN HIGH-SPEED MIXED-SIGNAL EMBEDDED SYSTEMS, AND PCB DESIGN, WITH HANDS-ON EXPERIENCE IN BOARD BRING-UP, VALIDATION, AND SYSTEM INTEGRATION

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Electrical and Computer Engineering, Applied Program – QPA: 3.95/4.00

May 2024

- Honors: Eta Kappa Nu (HKN) Honor Society
- Coursework: CubeSat Design-Build-Fly Lab, Space Robotics, Machine Learning for Signal Processing

Government College of Engineering and Research, Savitribai Phule Pune University

Pune, India

Bachelor of Engineering in Instrumentation and Control – GPA: 8.83/10.00

April 2020

PROFESSIONAL EXPERIENCE

Matic Robots

Mountain View, CA

February 2026 - Present

- **Electrical Design Engineer**
 - Design end-to-end PCBs for robotic subsystems with SoCs, MCUs.
 - Troubleshoot and root cause electrical issues on robots to find a short term and long term solutions.

Monarch Tractor

Livermore, CA

June 2024 – January 2026

- **Electrical Engineer**
 - Designed and validated **high-speed, multi-rail embedded compute boards**, including **MIPI/GMSL camera subsystems, power delivery, and sensor interfaces**; performed board bring-up, SI/PI analysis, and hardware debug using oscilloscopes, logic analyzers, and protocol analyzers.
 - Led the electrical engineering scope for **3+ OEM partnership projects** from concept to launch, owning system architecture and **collaborating with mechanical and firmware teams** on autonomous hardware bring-up, HV battery systems, OBC, and peripherals.
 - Extensive hands-on lab experience includes bring-up, debug, validation, and hardware test automation.
- **Intern (HV Battery Electrical System)**
 - Designed a **Battery Module Tester** for defect detection in Cell Modules before installation in the HV Battery Pack. Eliminated the need for two manual laborers and reduced inspection time by 80%

June 2023 - August 2023

Tantradnyan Engineers

Ahmednagar, India

April 2020 - November 2022

- **Project Engineer**
 - Led 3-member team; Cut production time by 20% through workflow optimization.

SKILLS

Hardware: High-speed PCB Design, Mixed-Signal Circuit Design, Schematic Capture, layout design; Microcontrollers (AVR, ST, ARM), sensor & camera subsystems; SerDes, Sensor Interfaces (ADC/DACs), PMICs, DC-DC converters, rail sequencing; Board Bring-Up, Validation Testing, Lab Test Equipment (scopes, digital analyzers etc.), Debugging

Software/Interfaces: Altium, KiCAD, LTSpice, High-speed (MIPI-CSI, GMSL), Low-speed (I2C, SPI, UART, CAN, JTAG), USB-C PD, Python scripting for validation tools, automated test workflows, and lab data processing

Programming Languages: Python, Embedded C; **Languages:** Fluent - English, Hindi; Native - Marathi

LEADERSHIP EXPERIENCE

Team Leader, Robotics Research Lab, Government College of Engineering and Research

Pune, India

- **Led 35-member robotics team** across **national competitions**, boosting productivity by 20%.
- Awarded with Special Jury Award among 80 participants teams in ABU Asia-Pacific Robot Contest 2019.

PUBLICATIONS

- "Design and Control of Omnidirectional Conveyor Model Using Image Processing," - IEEE
- "Design and Control of Quadruped Robot along with Machine Vision based Path Planning," – IEEE
- "Path Planning and Controlling of Omni-Directional Robot Using Cartesian Odometry and PID Algorithm," - IEEE

ACADEMIC PROJECTS

Smart Grid

September 2019 – May 2020

- Implemented Google Firebase cloud to control grid (mobile app and website); Used Allen Bradley PLC and SCADA

Rugby Ball pick-place and kick Robots

August 2019 – June 2020

Smart Street

January 2019 – March 2019

- Decreased energy usage up to 50%; Implemented vehicle speed-based retracting speed breaker.

Autonomous Quadruped Robot

August 2018 – June 2019

- Designed and built robot for ABU Robot Contest 2019 with vision-based path planning, obstacle avoidance

Pick and Place Automatic Guided Robot - Mahindra & Mahindra Engine Plant

November 2019 – December 2019

- Used high-resolution encoders and an IMU for position and orientation feedback

Calendar based Energy Management System - Mitsubishi Electric Cup

September 2018 – February 2019

- Developed Energy Management System for HVAC and Lighting: Decreased energy consumption by 10%